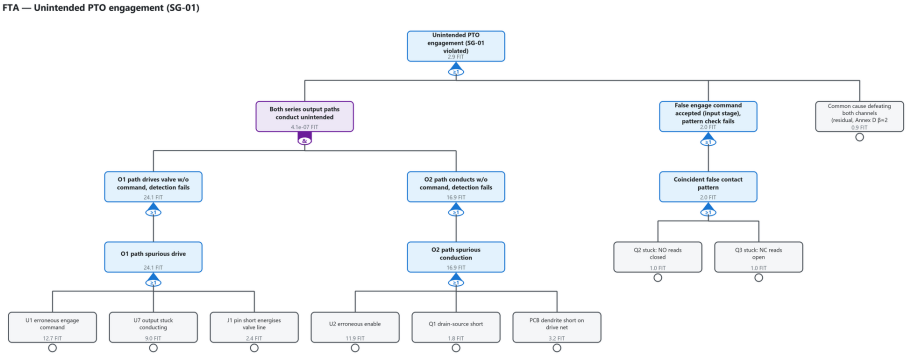
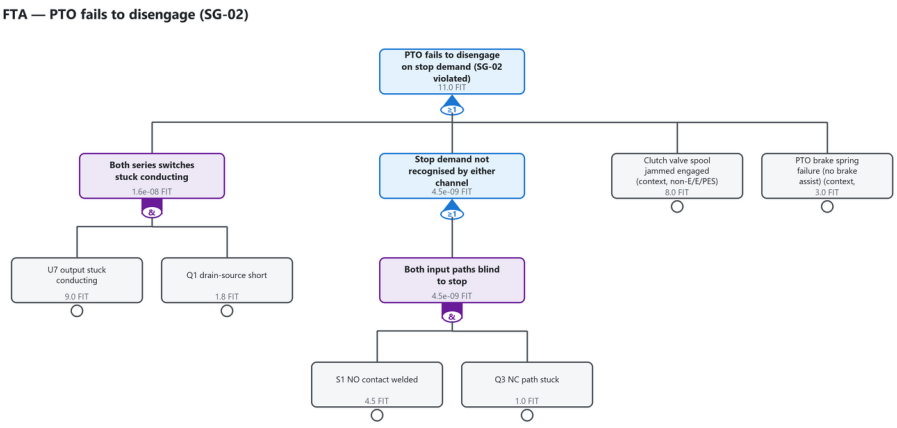
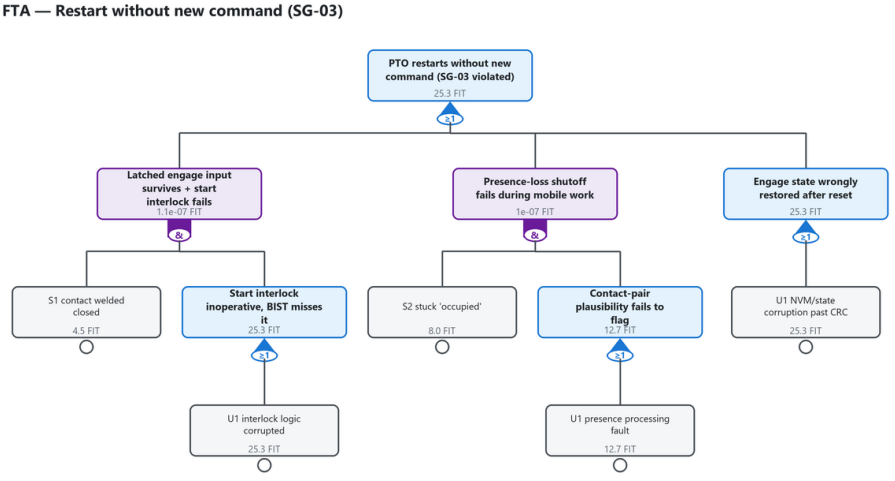
Analysis_Summary

ID	Contents	Artifact
PTO-ANA-000	**Safety analyses — summary and index** Quantitative analyses for the Category 3 / AgPL d verification, produced with the SafeSuite analysis toolchain and stored in this folder as native artifacts: SafeFIT reliability prediction (.sfit), SafeFMEA component FMEA (.sfmea) and SafeTree fault trees (.fta). All three share one BOM (rev C) and one mission profile, so every number is traceable to the same source. Open the artifacts with their editors; this document summarises the results and embeds the rendered trees.	—
PTO-ANA-001	**1. Reliability prediction (SafeFIT, IEC/TR 62380)** BOM rev C: 36 components, mission profile 'Agricultural tractor, cab ECU' (1 300 h/y powered, ΔT 33 K, 220 cycles/y). Predicted board failure rate λ = 570 FIT (= 0.0050 failures/year). Per-channel aggregation and the MTTFd conversion live in 4_Hardware/Reliability_Metrics with computed columns. Method notes: ICs computed per IEC/TR 62380 §7 with the thermal profile; electromechanical parts (switches, connectors) from the handbook tables with the T5 field history as sanity check; dangerous fraction per failure mode from the FMEA, not a blanket 50 %.	PTO_ECU_Reliability.sfit
PTO-ANA-002	**2. Component FMEA (SafeFMEA)** 67 failure-mode rows over 36 components; 28 modes are dangerous (safety-goal violating), of which 25 are controlled by a safety mechanism (D1–D5) with the diagnostic coverage recorded per row. Resulting DCavg (λ-weighted): 74.7 % (target ≥ 60 %, class medium). Residual single-point dangerous rate: 250.1 FIT. S/O/D rankings follow the project FMEA guide (S: 10 = entanglement fatality path, O from FIT deciles, D from diagnostic latency). Every dangerous-undetected mode appears as a basic event in the fault trees — the FMEA is the single source for mode ratios.	PTO_ECU_FMEA.sfmea
PTO-ANA-003	**3. FTA — SG-01 Unintended engagement**  *Top event residual 2.9 FIT — dominated by the CCF residual and the undetected fraction of the driver stages. The AND structure reflects the series O1·O2 output: both paths must fail conducting.* Mission time 8 760 h (one year, calendar basis of the FIT prediction). Basic-event rates are FMEA mode-ratio × SafeFIT component FIT; diagnosed events carry the FMEA mechanism coverage.	FTA_SG01_Unintended_Engagement.fta
PTO-ANA-004	**4. FTA — SG-02 Fails to disengage**  *Top event residual 11 FIT. The hydraulic spool-jam context event dominates; within the E/E/PES, both series switches must stick conducting — the architecture converts single stuck-on faults into availability events.*	FTA_SG02_Fail_To_Disengage.fta
PTO-ANA-005	**5. FTA — SG-03 Restart without command**  *Top event residual 25.3 FIT — double faults of latched input + failed interlock; the no-restore rule (PTO-TSR-005) removes every single-fault path.*	FTA_SG03_Auto_Restart.fta

ID	Contents	Artifact
PTO-ANA-006	**6. Consistency statement** Verified by the analysis pipeline (GenArtifacts self-check): every basic-event designator exists in the SafeFIT BOM; every FTA basic rate equals the FMEA mode share of that component's FIT; DCavg in Reliability_Metrics equals the FMEA aggregation; top-event numbers in this summary equal the .fta computations.	—